Python-Based LDA MCQs

1. Which Python library provides a built-in implementation of Linear Discriminant Analysis?
    - A) NumPy
    - B) Pandas
    - C) Scikit-learn
    - D) Matplotlib
2. Which class is commonly used to implement LDA in Scikit-learn?
    - A) PCA
    - B) LinearDiscriminantAnalysis
    - C) LogisticRegression
    - D) DecisionTreeClassifier
3. What is the purpose of the fit() method in Scikit-learn's LDA?
    - A) Visualize data
    - B) Learn discriminant directions from labeled data
    - C) Normalize features
    - D) Predict labels
4. Which method is used to reduce dimensionality after fitting an LDA model?
    - A) transform()
    - B) predict()
    - C) score()
    - D) classify()
5. In Python, which library is most suitable for computing eigenvalues and eigenvectors manually?
    - A) OpenCV
    - B) NumPy
    - C) Flask
    - D) Seaborn
6. What will happen if the target labels y are not provided while fitting an LDA model?
    - A) LDA will automatically cluster data
    - B) LDA cannot be trained properly
    - C) LDA converts to PCA

- D) No effect

7. Which command is used to import LDA from Scikit-learn?
   - A) from sklearn.decomposition import LDA
   - B) from sklearn.lda import LDA
   - C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
   - D) from sklearn.linear_model import LDA
8. What is the output shape after applying LDA with n_components=2 to a dataset containing 100 samples?
   - A) (2,100)
   - B) (100,2)
   - C) (100,100)
   - D) (2,2)
9. Which NumPy function is commonly used to compute eigenvalues and eigenvectors?
   - A) np.mean()
   - B) np.linalg.eig()
   - C) np.dot()
   - D) np.var()
10. In an LDA workflow, why is feature scaling often performed before applying LDA?
    - A) To reduce memory usage
    - B) To ensure features contribute comparably
    - C) To increase sample size
    - D) To remove labels
11. Which parameter controls the number of discriminant components in Scikit-learn LDA?
    - A) n_clusters
    - B) n_neighbors
    - C) n_components
    - D) max_depth
12. Which matrix operation is required when implementing LDA from scratch?
    - A) Matrix inversion
    - B) Image convolution
    - C) FFT
    - D) Hashing
13. While implementing LDA manually, which scatter matrix represents class compactness?

- A) Between-Class Scatter Matrix
- B) Within-Class Scatter Matrix
- C) Covariance Matrix
- D) Correlation Matrix

14. Which step comes first when implementing LDA from scratch?
    - A) Compute eigenvectors
    - B) Compute class means
    - C) Train classifier
    - D) Plot decision boundary
15. After computing eigenvectors in manual LDA, what should be done next?
    - A) Randomly shuffle them
    - B) Sort them according to eigenvalues
    - C) Delete negative values
    - D) Normalize labels
16. Which NumPy function can be used to project data onto selected discriminant vectors?
    - A) np.dot()
    - B) np.max()
    - C) np.sum()
    - D) np.argmax()
17. What is the primary purpose of lda.predict(X_test)?
    - A) Reduce dimensions
    - B) Classify unseen samples
    - C) Scale data
    - D) Compute covariance
18. Which metric is commonly used to evaluate an LDA classifier?
    - A) Accuracy
    - B) RMSE
    - C) MAPE
    - D) MAE
19. In Scikit-learn, LDA can be used for:
    - A) Classification only
    - B) Dimensionality reduction only

- C) Both classification and dimensionality reduction
- D) Clustering only

21. You are asked to implement Linear Discriminant Analysis from scratch using only NumPy (without using sklearn.discriminant_analysis.LinearDiscriminantAnalysis). Which sequence of steps is correct?

- A) Standardize Data → Compute Class Means → Compute Within-Class Scatter Matrix → Compute Between-Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$→ Select Top Eigenvectors → Project Data
- B) Compute FFT → Compute Covariance Matrix → Apply K-Means → Project Data
- C) Compute PCA Components → Apply Logistic Regression → Predict Labels
- D) Normalize Labels → Compute Decision Tree → Extract Features → Project Data

22. What is the purpose of the following line in a manual LDA implementation?

eigvals, eigvecs = np.linalg.eig(np.linalg.inv(Sw).dot(Sb))

- A) Compute principal components
- B) Compute discriminant directions
- C) Compute cluster centroids
- D) Compute feature scaling factors

23. Consider:

X_lda = X.dot(W)

What does W represent?

- A) Weight initialization matrix
- B) Matrix of selected discriminant vectors
- C) Covariance matrix
- D) Distance matrix

24. In a scratch implementation, why are eigenvectors sorted in descending order of eigenvalues?

- A) To reduce execution time
- B) To select directions with maximum discriminative power
- C) To normalize data
- D) To balance classes

25. When implementing LDA manually, what is the most likely issue if Sw is singular?

- A) Labels disappear
- B) Matrix inversion fails
- C) Data becomes nonlinear
- D) Eigenvalues become complex automatically